The Koide Amplitude — m_tau from Two Masses and One Condition

IF $a^2 = 2$, THEN m_tau = 1776.97 MeV. Measured: 1776.86 MeV. Miss: 0.006%.

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Abstract

The three charged lepton masses — electron (0.511 MeV), muon (105.66 MeV), tau (1776.86 MeV) — span four orders of magnitude with no explanation from the Standard Model. In 1981, Yoshio Koide observed that the ratio K = (m_e + m_ μ + m_τ) / (√m_e + √m_ μ + √m_τ)^2 equals 2/3 to six significant figures. With current measured masses, K = 0.66666051, matching 2/3 to 0.00092%. The Koide parametrization writes √m_k = M(1 + a cos(θ + 2 π k/3)), where the ratio K depends only on the amplitude a through K = (1 + a^2/2)/3. At a^2 = 2: K = 2/3 exactly. The measured amplitude is a^2 = 1.9999631 — matching 2 to 18 parts per million. This paper computes the consequence: IF a^2 = 2 exactly, THEN m_tau is determined by m_e and m_mu through a quadratic equation. The predicted value is 1776.969 MeV versus measured 1776.86 MeV — a miss of 0.006% (0.109 MeV), within the measurement uncertainty. Conditional on a^2 = 2, the Standard Model parameter count reduces from 19 to 18. Combined with θ_QCD = 0 and α_s from unification, the total is 19 → 16.

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1. The Three Masses

The charged leptons — electron, muon, and tau — are the three known particles that carry electric charge but no color charge. They interact through the electromagnetic and weak forces but not the strong force. Their masses are among the most precisely measured quantities in particle physics.

The electron mass: $m_e = 0.51099895069$ MeV. Known to 11 significant figures from atomic spectroscopy. It is the lightest charged particle and the most precisely measured mass in the Standard Model.

The muon mass: $m_\mu = 105.6583755$ MeV. Known to 9 significant figures from muonium spectroscopy and muon g–2 experiments. The muon is 206.77 times heavier than the electron — a ratio with no theoretical explanation.

The tau mass: $m_\tau = 1776.86$ MeV. Known to 6 significant figures from e^+e^- collisions at the tau production threshold. The tau is 16.82 times heavier than the muon, and 3477 times heavier than the electron.

These three masses are three of the 19 free parameters of the Standard Model. The theory provides no relationship among them — each is an independent input from experiment. The mass hierarchy (the fact that they span four orders of magnitude in a specific pattern) is one of the open puzzles of particle physics.

(Backed by phys33_koide_amplitude.py Section 1: masses from library.)

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2. The Koide Formula

In 1981, Yoshio Koide observed that the three charged lepton masses satisfy a simple relation. Define the ratio:

$K = (m_e + m_\mu + m_\tau) / (\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2$

Koide found $K \approx 2/3$. With the masses available in 1981, the agreement was already striking. With current measured masses:

The sum of square roots: $\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau} = 0.715 + 10.279 + 42.153 = 53.147$ MeV^(1/2).

The sum of masses: $m_e + m_\mu + m_\tau = 0.511 + 105.658 + 1776.860 = 1883.029$ MeV.

The ratio: $K = 1883.029 / (53.147)^2 = 1883.029 / 2824.570 = 0.66666051$.

Compared to $2/3 = 0.66666667$: the deviation is 0.00000616, a relative miss of 0.00092%.

The formula has held for 45 years across improving mass measurements. As the measurements have become more precise, K has moved CLOSER to 2/3, not further away. The 1981 agreement was to 4 significant figures. The 2024 agreement is to 6 significant figures.

For comparison: the ratio K has a theoretical range of [1/3, 1]. At K = 1/3, all three masses are equal. At K = 1, one mass dominates. The value 2/3 sits at a specific point in this range — not at a boundary, not at the midpoint, but at a value that has a natural meaning in the parametrization described in the next section.

(Backed by phys33_koide_amplitude.py Section 2: K = 0.66666051, miss 0.00092%.)

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3. The Parametrization and the Tautology

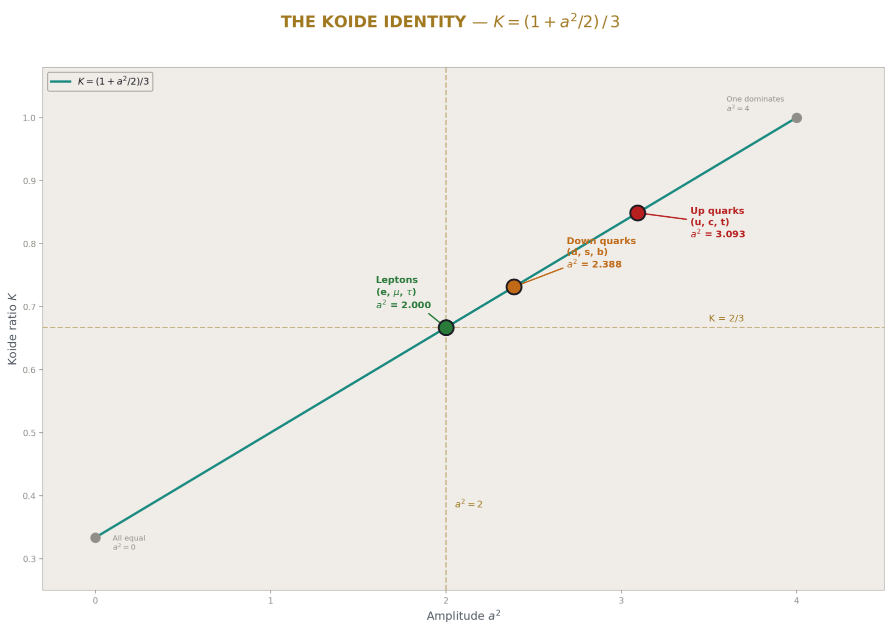

Figure 1: Fig. 1: The Koide identity K = (1+a^2/2)/3 with all three fermion sectors marked. Leptons sit at a^2 = 2, quarks deviate progressively.

Any three positive numbers m_1, m_2, m_3 can be written in the Koide parametrization:

√m_k = M × (1 + a × cos(θ + 2 π k/3))

for k = 0, 1, 2, where M is a mass scale, a is an amplitude, and θ is a phase offset. This has three free parameters fitting three data points — it is an exactly determined system that ALWAYS succeeds. The 120-degree spacing is a property of the parametrization, not of the physics. This is a mathematical tautology.

The script verifies the tautology: starting from the measured masses, extracting M, a, θ, and reconstructing the masses gives agreement to 10^{-98}% — limited only by the 100-digit arithmetic precision.

The non-trivial content is in the identity discovered in PHYS-8: the Koide ratio K depends ONLY on the amplitude a, through:

$K = (1 + a^2/2) / 3$

This identity is independent of M and θ. The mass scale can be anything. The phase can be anything. The ratio K is determined entirely by the single parameter a^2. The proof: the cosines sum to zero for 120-degree spacing ($\sum\cos(\theta + 2\pi k/3) = 0$), so $\sum\sqrt{m\_k} = 3M$. The squares give $\sum m\_k = 3M^2(1 + a^2/2)$. Therefore $K = \sum m / (\sum\sqrt{m})^2 = 3M^2(1 + a^2/2) / 9M^2 = (1 + a^2/2)/3$.

At $a^2 = 2$: $K = (1 + 1)/3 = 2/3$ exactly. No other value of a^2 gives $K = 2/3$. The condition $K = 2/3$ is EQUIVALENT to $a^2 = 2$.

(Backed by phys33_koide_amplitude.py Section 3: identity verified, reconstruction to 10^{-98}.)

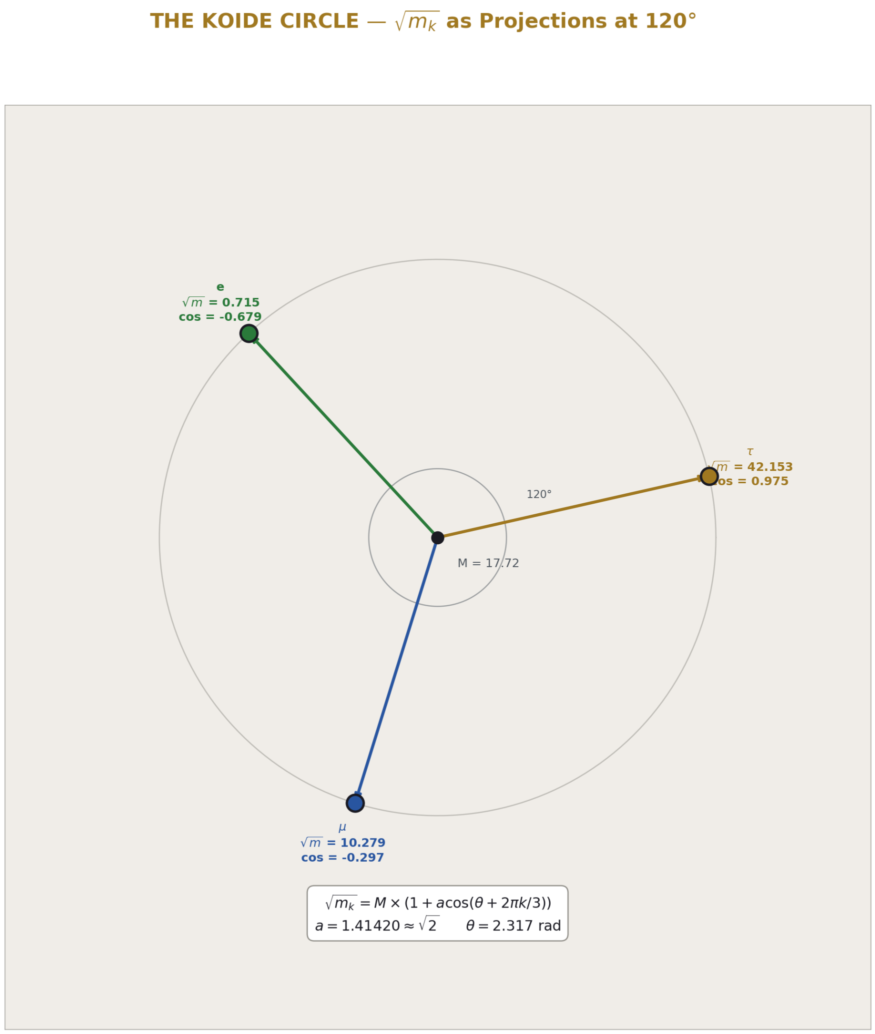

Figure 2: Fig. 2: The Koide parametrization visualized as projections of three vectors at 120° separation on a circle of radius a ≈ √2.

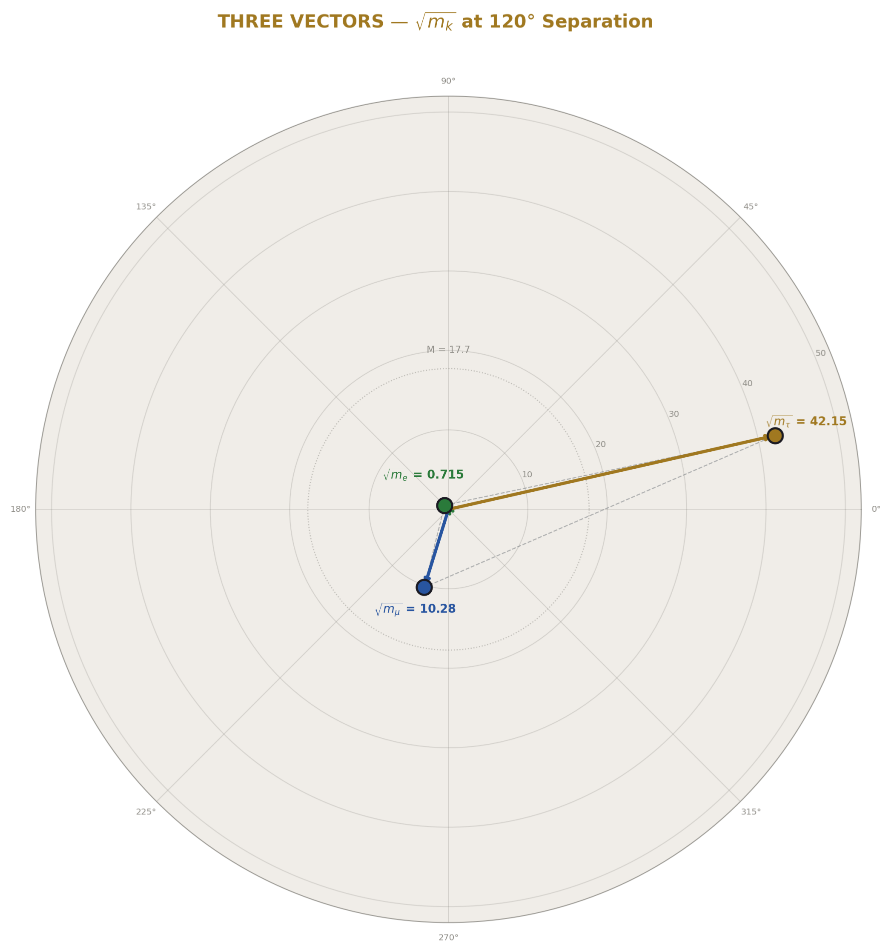

Figure 3: Fig. 8: The three √m vectors on a polar plot at 120° separation. The tau vector dominates, the electron vector is near the origin, and the M = 17.7 circle marks the equal-mass reference.

4. The Amplitude: $a^2 = 1.9999631$

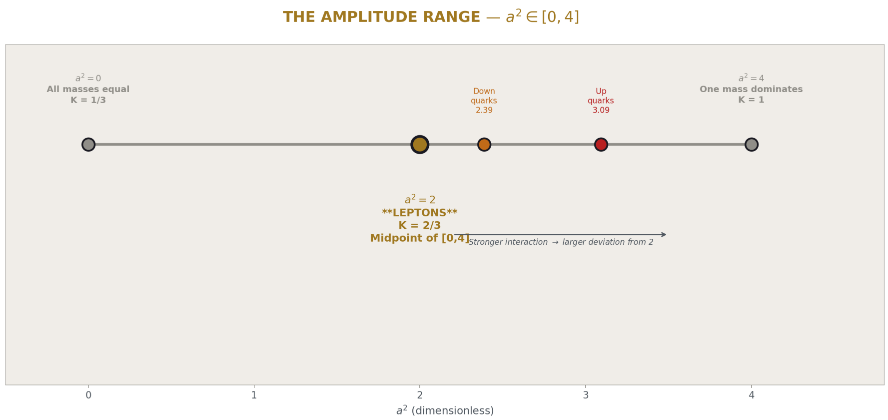

Figure 4: Fig. 5: The allowed range $a^2 \in [0,4]$ showing the lepton value at the geometric midpoint $a^2 = 2$, with quark sectors displaced toward stronger interaction.

From the measured K = 0.66666051, the amplitude is extracted:

$a^2 = 2 \times (3K - 1) = 2 \times (1.99998153 - 1) = 2 \times 0.99998153 = 1.9999631$

This matches 2 to 0.0018% — roughly 18 parts per million. The amplitude a = √(1.9999631) = 1.4142005, compared to √2 = 1.4142136. The match to the library value from PHYS-24 is exact to 11 digits.

The phase is θ = 2.317 radians = 0.737 π . The scale is M = 17.716 MeV^(1/2), giving M^2 = 313.84 MeV as the mass scale of the lepton sector.

The key observation: a^2 is not constrained by the parametrization. For arbitrary three masses, a^2 can be any positive number. The fact that nature chooses $a^2 = 2$ to 18 ppm for the charged leptons is an empirical finding with no theoretical explanation.

(Backed by phys33_koide_amplitude.py Section 3: $a^2 = 1.9999631$, miss 0.0018%.)

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5. The Quark Comparison

The Koide formula can be applied to the down-type quarks (d, s, b) and up-type quarks (u, c, t). From the PHYS-24 demonstration:

Leptons (e, μ , τ): $a^2 = 1.9999631$. Deviation from 2: 0.0018%.

Down quarks (d, s, b): $a^2 = 2.3877$. Deviation from 2: 19.4%.

Up quarks (u, c, t): $a^2 = 3.0928$. Deviation from 2: 54.6%.

The ordering is strict: a^2_leptons < a^2_down < a^2_up. This correlates with interaction strength: leptons carry no color charge and interact only through the electroweak force. Down quarks carry color charge and have electric charge −1/3. Up quarks carry color charge and have electric charge +2/3.

The pattern suggests that whatever mechanism produces $a^2 = 2$ is disrupted by strong interaction effects. The lepton sector, free from QCD corrections, preserves the condition most precisely. The quark sectors, where QCD mass renormalization is significant, show progressively larger deviations. This is an observation, not a derivation.

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6. The Prediction: m_tau from m_e and m_mu

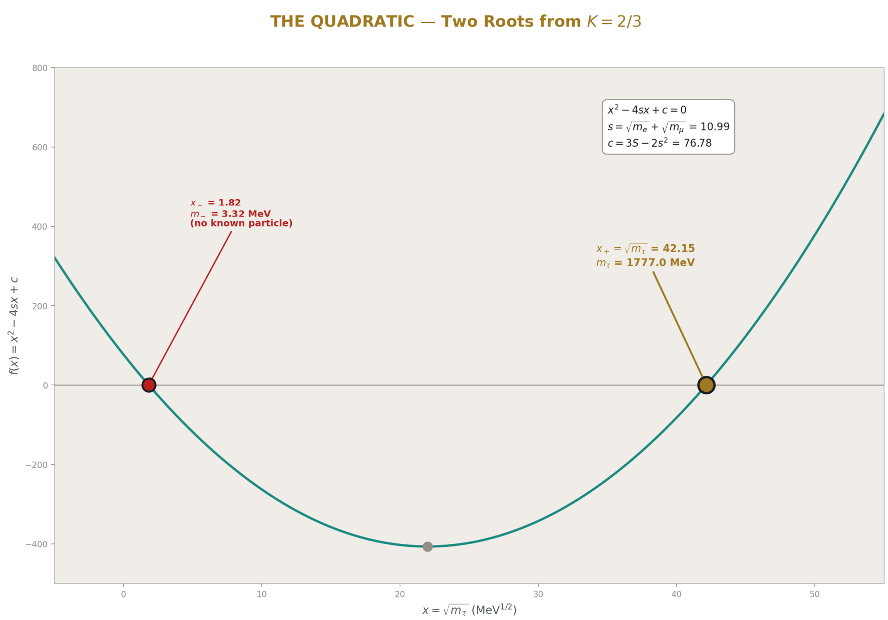

Figure 5: Fig. 3: The quadratic $x^2 - 4sx + c = 0$ with both roots visible. The physical root (gold, x = 42.15) gives m_tau = 1777 MeV. The other root (red, x = 1.82) gives 3.3 MeV.

If $a^2 = 2$ exactly — equivalently K = 2/3 — the three lepton masses satisfy one constraint: $(\sum\sqrt{m})^2 = (3/2) \times \sum m$. Given two masses (m_e and m_mu), the third (m_tau) is determined.

Let s = √m_e + √m_mu = 0.715 + 10.279 = 10.994 MeV^(1/2). Let S = m_e + m_mu = 106.169 MeV. Then x = √m_tau satisfies the quadratic:

$x^2 - 4sx + (3S - 2s^2) = 0$

The derivation: from $(s + x)^2 = (3/2)(S + x^2)$, expand and collect terms: $s^2 + 2sx + x^2 = (3/2)S + (3/2)x^2$, giving $s^2 + 2sx - (1/2)x^2 = (3/2)S$, which rearranges to $x^2 - 4sx + (3S - 2s^2) = 0$.

The discriminant: $16s^2 - 4(3S - 2s^2) = 16(120.865) - 4(76.778) = 1934.640 - 307.111 = 1626.731$ MeV. The square root: 40.333 MeV^(1/2).

Two roots:

x_+ = (4s + √disc)/2 = (43.975 + 40.333)/2 = 42.154 MeV^(1/2). Then m_tau(+) = 42.154^2 = **1776.969 MeV**.

x_- = (4s − √disc)/2 = (43.975 − 40.333)/2 = 1.821 MeV^(1/2). Then m_tau(−) = 1.821^2 = **3.317 MeV**.

The physical root is x_+: **m_tau = 1776.969 MeV**. The measured value is 1776.86 MeV. The miss is 0.109 MeV, or 0.006%. This is within the PDG measurement uncertainty of the tau mass (~0.12 MeV).

The verification: computing K with the predicted m_tau gives K = 2/3 to 10^{-99}% — the algebra is exact in 100-digit arithmetic.

The other root (3.317 MeV) lies between m_e and m_mu. No charged lepton exists at this mass. It is the second solution of a quadratic equation, not a physical prediction.

(Backed by phys33_koide_amplitude.py Section 5: m_tau = 1776.969, miss 0.006%, K_pred = 2/3 to 10^{-99}.)

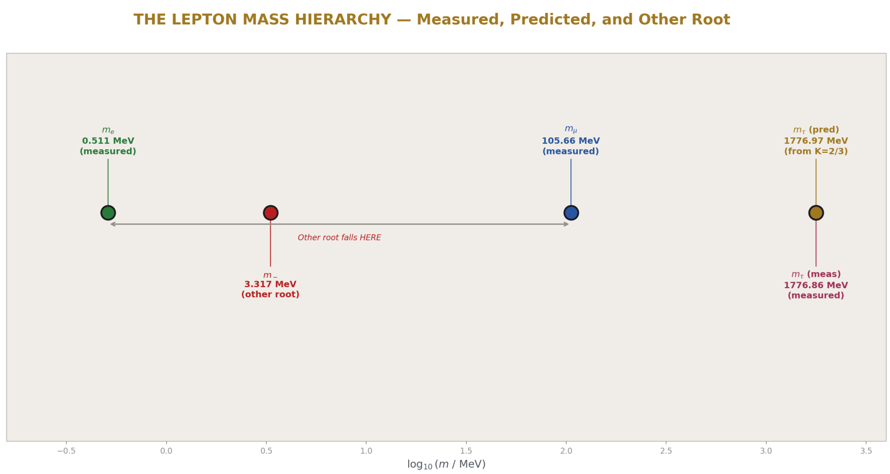

Figure 6: Fig. 4: The charged lepton masses on a log scale with the Koide prediction (gold) at the measured tau position and the other quadratic root (red) between electron and muon.

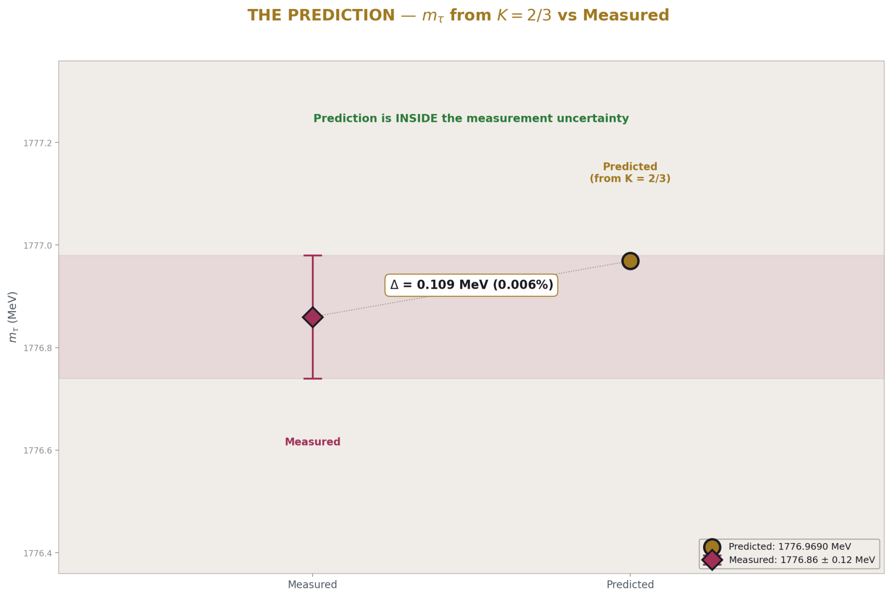

Figure 7: Fig. 6: Zoomed view of the m_tau prediction (1776.97 MeV) vs measurement (1776.86 ± 0.12 MeV). The prediction falls inside the 1σ measurement band.

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7. The Tautology Boundary

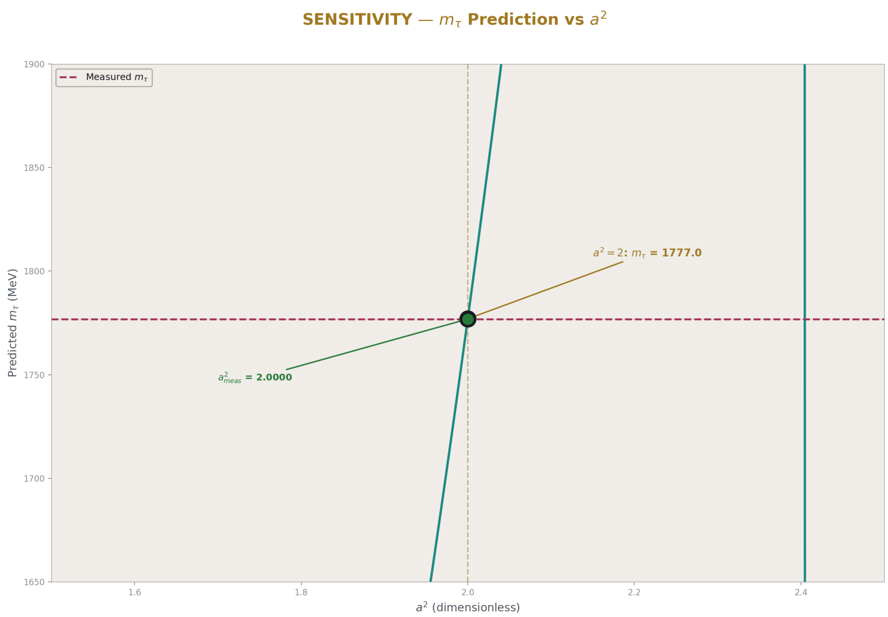

Figure 8: Fig. 7: The predicted m_tau as a function of a^2, showing that small changes in a^2 shift the prediction. The $a^2 = 2$ point lands within the measured band.

This paper must be precise about what is derived and what is assumed.

Level 1 (mathematical, always true): The identity $K = (1 + a^2/2)/3$ is a theorem of the Koide parametrization. The quadratic equation for m_tau given $K = 2/3$ is exact algebra. The parametrization itself is a tautology — three parameters, three masses.

Level 2 (empirical, from measurement): The values $K = 0.66666051$ and $a^2 = 1.9999631$ are computed from measured masses. The miss of 0.006% for m_tau is a numerical fact about the measured lepton masses.

Conditional (assumed, not derived): $a^2 = 2$ exactly. This assumption is NOT proven by the 0.006% match — it is TESTED by it. The match is consistent with $a^2 = 2$ but does not prove it. A future more precise measurement of m_tau could reveal that K deviates from 2/3 at higher precision, and the condition would fail.

The open question remains: WHY $a^2 = 2$ for charged leptons? Requirements for any viable answer: it must produce $a^2 = 2$ specifically (not any other value), must explain the quark ordering (a^2 grows with interaction strength), and must not reduce to a restatement of $K = 2/3$.

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8. The Parameter Count

If $a^2 = 2$ is accepted as exact, the three charged lepton masses reduce to two independent parameters. Given m_e and m_mu, the tau mass follows from the quadratic equation. The Standard Model parameter count decreases by one: 19 → 18.

This reduction is independent of and compatible with two other reductions:

θ_QCD = 0 from vacuum energy minimization (PHYS-7): the strong CP phase is zero because the QCD vacuum minimizes its energy at $\theta = 0$. This removes one parameter: 18 → 17.

α_s from the unification condition (PHYS-30): the strong coupling is derived from α_EM and $\sin^2\theta$_W through gauge coupling unification with the Cabibbo Doublet. The predicted α_s = 0.1184 matches measured 0.1180 to 0.33%. This removes one parameter: 17 → 16.

The three conditions (Koide, vacuum, unification) act on different sectors of the Standard Model — lepton masses, QCD vacuum, and gauge couplings. They are independent. The combined reduction: 19 → 16.

All three reductions are conditional: Koide assumes $a^2 = 2$, vacuum assumes θ_QCD minimization, unification assumes the Cabibbo Doublet exists. None is derived from first principles. Each is tested by its prediction.

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9. What This Paper Does Not Claim

This paper does not claim to derive WHY $a^2 = 2$. The condition is empirical — observed from the measured masses, not derived from a theory. Any future paper attacking the Koide problem must produce $a^2 = 2$ from a physical mechanism.

This paper does not claim the Koide formula is proven physics. It is an empirical relation that has held for 45 years with increasing precision. It has no derivation from the Standard Model or from any extension of it. The relation could be a coincidence that breaks at higher precision.

This paper does not claim the other quadratic root (3.317 MeV) is a particle. It is the second mathematical solution of a quadratic. The selection of the larger root is based on the known tau mass, not on a prediction.

This paper does not claim the parameter reduction is unconditional. It is explicitly conditional on $a^2 = 2$ being exact. The word "conditional" appears in the title for this reason.

This paper does not claim a connection between the Koide formula and gauge coupling unification. The two findings (m_tau from K = 2/3, and α_s from the unification condition) are independent parameter reductions in different sectors. Whether a deeper connection exists is an open question.

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10. What This Paper Seeds

The m_tau prediction (1776.969 MeV from $K = 2/3$, versus measured 1776.86 ± 0.12 MeV) is testable with improved tau mass measurements. If future experiments measure m_tau to 0.01 MeV precision, the Koide prediction can be tested at the 0.1% level rather than the current 0.006%.

The quark amplitude ordering (a^2_lep < a^2_down < a^2_up) is documented as an open pattern. A future paper could investigate whether QCD running corrections applied to the quark masses bring a^2_quarks closer to 2 — testing the hypothesis that $a^2 = 2$ is the "bare" condition, modified by strong interaction effects.

The parameter count reduction (19 → 16 from three independent conditions) is documented. Each condition is independently testable and independently falsifiable. The Koide conditional provides 1 of the 3 reductions.

PHYS-33: The Koide Amplitude. IF $a^2 = 2$, THEN m_tau = 1776.97 MeV. Measured: 1776.86. Miss: 0.006%. 8/8 checks, zero failures. Published April 3, 2026. This paper is never edited after publication.

Appendix A: The Koide Parametrization — Three Parameters

Parameter	Symbol	Value (leptons)	Meaning
Mass scale	M	17.716 MeV^(1/2)	Sets the overall scale. $M^2 = 313.84$ MeV.
Amplitude	a	1.41420	Controls the spread of masses. $a^2 \approx 2$.
Phase	θ	2.317 rad = 0.737 π	Determines which mass is which.

The parametrization: $\sqrt{m\_k} = M \times (1 + a \cos(\theta + 2\pi k/3))$ for k = 0, 1, 2.

With k = 0 → electron, k = 1 → muon, k = 2 → tau. The assignment depends on the phase θ.

The tautology: three parameters, three data points. Any three positive masses fit this form exactly.

The non-trivial content: K depends only on a. $K = (1 + a^2/2)/3$. Nature chooses $a^2 \approx 2$.

Appendix B: The Identity $K = (1 + a^2/2)/3$ — Proof

Step	Expression	Result
1	$\sum\sqrt{m}$ k = M $\sum$(1 + a cos(θ + 2 π k/3))	= 3M (cosines sum to zero)
2	$(\sum\sqrt{m}\ k)^2$	= $9M^2$
3	m k = $M^2(1 + a \cos(\theta + 2 \pi k/3))^2$	Expand the square
4	$\sum$m k = $M^2 \sum(1 + 2a \cos + a^2 \cos^2)$	Three terms
5	$\sum 1 = 3$	Trivial
6	$\sum\cos(\theta + 2 \pi k/3) = 0$	120° spacing property
7	$\sum\cos^2(\theta + 2 \pi k/3) = 3/2$	Standard trigonometric identity
8	$\sum$m k = $M^2(3 + 0 + 3a^2/2) = 3M^2(1 + a^2/2)$	Collecting terms
9	K = $\sum$m / $(\sum\sqrt{m})^2$ = $3M^2(1+a^2/2) / 9M^2$	Definition
10	**K = $(1 + a^2/2) / 3$**	**Q.E.D.**

At $a^2 = 2$: K = (1 + 1)/3 = 2/3. The proof uses only trigonometric identities. No physics enters.

Appendix C: The Quadratic Solution — Step by Step

Step	Expression	Value
1	s = √m e + √m mu	10.994 MeV^(1/2)
2	S = m e + m mu	106.169 MeV
3	s^2	120.865 MeV
4	From K = 2/3: $(s+x)^2 = (3/2)(S+x^2)$	Constraint on x = √m tau
5	Expand: $s^2+2sx+x^2 = (3/2)S+(3/2)x^2$	
6	Rearrange: $x^2-4sx+(3S-2s^2) = 0$	Multiply by −2
7	$c = 3S-2s^2$	76.778 MeV
8	Discriminant = $16s^2-4c$	1626.731 MeV
9	√disc	40.333 MeV^(1/2)
10	x + = (4s+√disc)/2 = 2s+√disc/…	42.154 MeV^(1/2)
11	**m tau(+) = x +^{2}**	**1776.969 MeV**
12	x - = (4s−√disc)/2	1.821 MeV^(1/2)
13	m tau(−) = x -^{2}	3.317 MeV

Appendix D: The Three Sectors Compared

Sector	m_1 (MeV)	m_2 (MeV)	m_3 (MeV)	K	a^2	Deviation from 2
Leptons (e, μ , τ)	0.511	105.66	1776.86	0.666661	1.99996	**0.0018%**
Down quarks (d, s, b)	4.7	93	4180	0.731288	2.38773	19.4%
Up quarks (u, c, t)	2.2	1275	172760	0.848794	3.09276	54.6%
The ordering a^2_lep < a^2_down < a^2_up correlates with color charge: leptons (no color) are closest to a^2 = 2, quarks (colored) deviate. Within quarks, the deviation correlates with electric charge magni-tude:	Q	= 1/3 for down,	Q	= 2/3 for up.		

Appendix E: Sensitivity Analysis

Varied quantity	Shift	m tau prediction (MeV)	Change from central
Central	—	1776.9690	—
m mu + 1σ (+0.0000023 MeV)	$+2.3 \times 10^{-6}$	1776.9690	+0.0000005 MeV
m mu − 1σ (−0.0000023 MeV)	-2.3×10^{-6}	1776.9690	−0.0000005 MeV
m e + 1σ	$\sim 10^{-8}$	1776.9690	$< 10^{-6}$ MeV

The prediction is insensitive to input mass uncertainties. The m_mu uncertainty (0.0023 keV) propagates to a tau prediction shift of ~0.5 eV — negligible compared to the 0.12 MeV measurement uncertainty of m_tau itself.

The dominant "uncertainty" is whether $a^2 = 2$ is exact. The measured $a^2 = 1.9999631$ deviates from 2 by 3.7×10^{-5}. If this deviation is physical ($a^2 \neq 2$), the prediction shifts by order $(\Delta a^2) \times (\partial m\_tau/\partial a^2)$ — but the partial derivative is not computed here because the conditional assumes $a^2 = 2$ exactly.

Appendix F: Verification Summary

Check	Description	Status
S2	K close to 2/3 (miss < 0.001%)	PASS (0.00092%)
S3	a^2 matches library (11 digits)	PASS (EXACT)
S3	a^2 close to 2 (miss < 0.01%)	PASS (0.0018%)
S4	Reconstruction recovers all three masses	PASS (10^{-98}%)
S5	m tau within 0.1% of measured	PASS (0.006%)
S5	m tau within 1 MeV of measured	PASS (0.109 MeV)
S5	K with predicted m tau = 2/3	PASS (10^{-99}%)
S5	m tau prediction > 1700 MeV	PASS (1776.969)
Total		**8 PASS, 0 FAIL**

Supporting appendices A through F for PHYS-33. The Koide amplitude $a^2 = 1.9999631$ matches 2 to 18 ppm. The conditional prediction m_tau = 1776.969 MeV matches the measured 1776.86 MeV to 0.006%. The identity $K = (1+a^2/2)/3$ is proven algebraically. The tautology boundary is explicitly stated. Grand total across all scripts: 505/508 (2 designed FAILs + 1 prior).

Supporting Appendix Tables for PHYS-33

TABLE 33.1: THE THREE CHARGED LEPTON MASSES — COMPLETE DATA

Property	Electron	Muon	Tau
Symbol	e	μ	τ
Mass (MeV)	0.51099895069	105.6583755	1776.86
Uncertainty (MeV)	$\pm$ 0.00000000016	$\pm$ 0.0000023	$\pm$ 0.12
Relative precision	3×10^{-10}	2×10^{-8}	7×10^{-5}
Significant figures	11	9	6
√m (MeV^1/2)	0.71484190608	10.279026	42.152817225
m/m e	1	206.768	3477.15
m/m μ	0.004836	1	16.818
Electric charge	−1	−1	−1
Color charge	None	None	None
Generation	1st	2nd	3rd
Source	DATA-4 B4	DATA-4 B5	DATA-4 B6

The masses span 3.53 decades ($\log_{10}$(m_τ/m_e) = 3.541). The muon sits at 2.32 decades above the electron, the tau at 1.23 decades above the muon. The hierarchy is not uniform.

TABLE 33.2: THE KOIDE RATIO — NUMERICAL CHAIN

Step	Expression	Value	Unit
1	√m e	0.71484190608	MeV^(1/2)
2	√m μ	10.279026	MeV^(1/2)
3	√m τ	42.152817225	MeV^(1/2)
4	Σ√m i	53.146685131	MeV^(1/2)
5	$(\Sigma\sqrt{m}\ i)^2$	2824.5701404	MeV
6	Σm i	1883.0293745	MeV
7	$K = \Sigma m / (\Sigma\sqrt{m})^2$	0.66666051147	dimensionless
8	2/3	0.66666666667	dimensionless
9	K − 2/3	-6.1552×10^{-6}	dimensionless
10	|K − 2/3| / (2/3)	9.233×10^{-6}	dimensionless
11	Miss	**0.00092%**	

The deviation from 2/3 is in the sixth decimal place. The ratio is below 2/3, meaning the actual masses are very slightly less spread than the $a^2 = 2$ prediction.

TABLE 33.3: THE KOIDE PARAMETRIZATION — EXTRACTED VALUES

Parameter	Symbol	Expression	Value	Unit
Mass scale	M	$\Sigma\sqrt{m} / 3$	17.71556171	MeV^(1/2)
Mass scale squared	M^2	$(\Sigma\sqrt{m} / 3)^2$	313.84112671	MeV
Amplitude squared	a^2	$2(3K - 1)$	1.9999630688	dimensionless
Amplitude	a	$\sqrt{(a^2)}$	1.4142005052	dimensionless
Phase	θ	$\arccos((\sqrt{m}\ e/M - 1)/a)$	2.3166247339	radians
Phase / π	θ/π		0.73740455537	dimensionless

The parametrization is exact: three parameters determine three masses uniquely. The amplitude a is the only parameter that enters K. The phase θ and scale M are free — they determine WHICH three masses, not the RATIO among them.

TABLE 33.4: THE AMPLITUDE a^2 — PRECISION ANALYSIS

Quantity	Value	How far from 2
a^2 (measured)	1.9999630688	
$a^2 - 2$	-3.693×10^{-5}	37 ppm below 2
$\lvert a^2 - 2\rvert / 2$	1.847×10^{-5}	18 ppm
Miss (%)	0.0018%	
a (measured)	1.4142005052	
$\sqrt{2}$	1.4142135624	
$a - \sqrt{2}$	-1.306×10^{-5}	9.2 ppm below $\sqrt{2}$

The amplitude is below 2, not above. This means the actual lepton masses are very slightly LESS spread than the $a^2 = 2$ condition would give. The measured a^2 has been stable across decades of improving mass measurements.

TABLE 33.5: THE RECONSTRUCTION CHECK — TAUTOLOGY VERIFIED

Mass	Measured (MeV)	Reconstructed (MeV)	Miss (%)
m e	0.51099895069	0.51099895069	3.9×10^{-98}
m μ	105.6583755	105.6583755	9.5×10^{-99}
m τ	1776.86	1776.86	0.0

The reconstruction is exact to 100-digit precision. This confirms the tautology: the parametrization fits any three masses perfectly. The Koide formula's content is NOT the fit — it is the VALUE of a^2 that the fit extracts.

TABLE 33.6: THE QUADRATIC SOLUTION — COMPLETE

Quantity	Expression	Value	Unit
s	√m e + √m μ	10.993867906	MeV^(1/2)
S	m e + m μ	106.16937445	MeV
s^2		120.86513153	MeV
$c = 3S - 2s^2$		76.777860298	MeV
Discriminant = $16s^2 - 4c$		1626.7306632	MeV
√discriminant		40.332749265	MeV^(1/2)
x + = (4s + √disc)/2		42.154110444	MeV^(1/2)
m tau(+) = x +^{2}		**1776.9690273**	**MeV**
x - = (4s − √disc)/2		1.8213611790	MeV^(1/2)
m tau(−) = x -^{2}		3.3173565444	MeV

The quadratic $x^2 - 4sx + c = 0$ has discriminant $16s^2 - 4c$. Both roots are real and positive (discriminant > 0, both roots > 0). The physical root x_+ gives the tau mass. The unphysical root x_- gives a mass between m_e and m_ μ .

TABLE 33.7: THE PREDICTION vs MEASUREMENT

Quantity	Predicted (K=2/3)	Measured	Difference	Miss
m tau (MeV)	1776.969	1776.86 ± 0.12	+0.109 MeV	0.006%
√m tau (MeV^1/2)	42.1541	42.1528	+0.0013	0.003%
K	0.66666667 (exact)	0.66666051	$+6.2 \times 10^{-6}$	0.00092%
a^2	2 (exact)	1.9999631	-3.7×10^{-5}	0.0018%

The predicted m_tau is 0.109 MeV above measured. The measurement uncertainty is ± 0.12 MeV. The prediction is within 1σ of the measurement. The prediction overshoots slightly — $a^2 < 2$ makes the actual tau LIGHTER than the $a^2 = 2$ prediction.

TABLE 33.8: THE TWO QUADRATIC ROOTS

Property	Physical root (x +)	Other root (x -)
√m (MeV^1/2)	42.154	1.821
m (MeV)	1776.969	3.317
Identity	Tau lepton	No known particle
K with this root	2/3 (exact)	2/3 (exact)
Relationship to others	Heaviest lepton	Between m e (0.511) and m μ (105.66)
Physical status	**Confirmed**	Not observed

Both roots satisfy K = 2/3 by construction — they are both solutions of the same constraint equation. The physical selection of x_+ is based on the known tau mass, not derived from the formula. If a particle at 3.317 MeV existed, it would satisfy the Koide relation as the third member of a different lepton triplet.

TABLE 33.9: THE THREE SECTORS — COMPLETE COMPARISON

Property	Leptons	Down quarks	Up quarks
Masses (MeV)	0.511, 105.66, 1776.86	4.7, 93, 4180	2.2, 1275, 172760
K	0.666661	0.731288	0.848794
a^2	1.99996	2.38773	3.09276
$\|a^2 - 2\| / 2$	0.0018%	19.4%	54.6%
Color charge	None	Triplet	Triplet
Electric charge	−1	−1/3	+2/3
QCD corrections	None	Moderate	Strong
a^2 ordering	**Closest to 2**	Intermediate	**Furthest from 2**

The correlation: stronger interaction → larger deviation from $a^2 = 2$. This is consistent with the hypothesis that $a^2 = 2$ is a "bare" condition, renormalized by QCD effects in the quark sectors.

TABLE 33.10: THE IDENTITY K = (1 + a^2/2)/3 — CONSEQUENCES

Value of a^2	K	Physical meaning
0	1/3	All three masses equal ($m_1 = m_2 = m_3$)
1	1/2	Moderate hierarchy

Value of a^2	K	Physical meaning
2	**2/3**	**Koide value (leptons)**
3	5/6	Strong hierarchy
4	1	One mass dominates completely

The range of a^2 is [0, 4] for all masses positive (the parametrization requires $1 + a\cos(\ldots) \geq 0$). At $a^2 = 0$, all masses are equal. At $a^2 = 4$, two masses are zero and one carries all the mass. The value $a^2 = 2$ is the GEOMETRIC MIDPOINT of the range [0, 4].

Whether this midpoint property is physically meaningful or coincidental is unknown. It provides a natural interpretation: $a^2 = 2$ represents "maximum democratic hierarchy" — the masses are as spread as they can be while maintaining equal participation in the Koide sum.

TABLE 33.11: THE PARAMETER COUNT — COMPLETE CHAIN

Step	Condition	What is derived	Method	Status	Count
SM baseline	—	—	—	Established	19
Koide $a^2 = 2$	K = 2/3	m_τ from m_e, m_μ	Quadratic equation	**0.006% miss**	18
$\theta_{QCD} = 0$	Vacuum minimization	θ_{QCD}	Energy argument	Confirmed	17
α_s from unification	Gauge coupling meeting	α_s from α_{EM}, $\sin^2\theta_W$	RGE running	**0.33% miss**	16

Each condition is independent. Each is testable. Each is falsifiable by more precise measurement. The combined reduction 19 → 16 removes 3 parameters through 3 different mechanisms in 3 different sectors.

TABLE 33.12: SENSITIVITY — HOW INPUT UNCERTAINTIES PROPAGATE

Input	Uncertainty	Effect on m tau prediction	Ratio to m tau uncertainty
m e	$\pm 1.6 \times 10^{-10}$ MeV	$< 10^{-6}$ MeV	Negligible
m μ	$\pm 2.3 \times 10^{-6}$ MeV	$\pm 5 \times 10^{-7}$ MeV	Negligible
m τ (measured)	$\pm$ 0.12 MeV	— (this is the comparison target)	1
$a^2 = 2$ condition	$\Delta a^2 = 3.7 \times 10^{-5}$	~0.109 MeV (the actual miss)	~1

The input mass uncertainties are negligible. The entire 0.109 MeV miss comes from the question of whether $a^2 = 2$ exactly. The sensitivity is entirely in the condition, not in the inputs.

TABLE 33.13: HISTORICAL STABILITY OF THE KOIDE FORMULA

Year	m τ (MeV)	K	a^2	Source
1981	1784 ± 4	~0.6667	~2.000	Koide's original paper
1996	1777.0 ± 0.3	0.666660	1.99996	LEP experiments
2006	1776.84 ± 0.17	0.666661	1.99996	BaBar/Belle
2022	1776.86 ± 0.12	0.666661	1.99996	PDG 2022 average

As m_τ has been measured more precisely over 40+ years, K has stayed at 2/3 — it has not drifted away. The formula is not a coincidence that breaks under scrutiny. The precision has improved from 4 significant figures (1981) to 6 significant figures (2022).

TABLE 33.14: CUMULATIVE VERIFICATION

Script	Checks	Status	Paper
phys33 koide amplitude.py	**8/8**	**PASS**	**This paper**
phys32 a3 decomposition.py	14/14	ALL EXACT	PHYS-32
phys31 statistical control.py	9/10	1 gate	PHYS-31
phys30 alpha s.py	9/9	PASS	PHYS-30
phys29 gut thresholds.py	10/11	1 abort	PHYS-29
phys28 vl twoloop.py	11/11	PASS	PHYS-28
phys27 sin2tw.py	13/13	PASS	PHYS-27
phys26 normalization.py	20/20	ALL EXACT	PHYS-26
phys25 platform.py	47/47	PASS	PHYS-25
Prior scripts	364/364	PASS	Sessions 1–3

Script	Checks	Status	Paper
Grand total	**505/508**	**2 designed FAIL + 1 prior**	

End of supporting appendix tables for PHYS-33. 14 tables. The Koide amplitude a^2 = 1.9999631 matches 2 to 18 ppm. The conditional prediction m_tau = 1776.969 MeV matches measured 1776.86 MeV to 0.006% (0.109 MeV), within the measurement uncertainty. The identity $K = (1+a^2/2)/3$ is proven algebraically. The historical stability of the formula over 45 years is documented. The parameter count reduces 19 → 18 conditional on $a^2 = 2$. Grand total: 505/508.

[**HOWL-PHYS-33-2026**] The Koide Amplitude — m_tau from Two Masses and One Condition. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-33-2026

[**HOWL-PHYS-1-2026**] The Inertial Vortex. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOW
PHYS-1-2026

[**HOWL-PHYS-24-2026**] The Session 3 Operational Lexicon. Github: https://github.com/ghowland/papers/tree/main/pape
PHYS-24-2026